

AERIS

AUTONOMOUS ENTERPRISE TRUST, NEGOTIATION & COORDINATION LAYER

PRODUCT MANAGEMENT CASE STUDY

AI-Agent Enterprise Procurement • Decision Intelligence • Zero-Trust Coordination

ROLE

Product Manager / Technical Product Owner

PROJECT TYPE

Distributed Multi-Agent AI + Zero-Trust Architecture

CORE JOURNEY

Telemetry → RFQ → M2M → AHP → Vault → Settlement

PRODUCT PROMISE

Make autonomous machine-to-machine procurement and settlement transparent, explainable, cryptographically verifiable, and audit-ready.



1. Executive Summary

AERIS is an enterprise coordination and trust architecture designed to solve a fundamental barrier in autonomous supply chains: how distributed machines, agents, and suppliers negotiate contracts, establish trust, and verify settlements without centralized intermediaries. The prototype combines real-time 3D global telemetry, multi-agent negotiation, multi-criteria decision modeling, and cryptographic ledger immutability.

The Product Management challenge is not merely building a static dashboard or a basic blockchain mock-up. It is designing a verifiable decision layer that models complex trade-offs across price, lead time, ESG compliance and reliability, enforces zero-trust auditability on demand, and makes telemetry understandable.

2. Product Context

Global procurement workflows depend on fragmented ERPs, manual vendor negotiations, opaque supplier scorecards, and slow post-delivery reconciliation. AERIS establishes a standardized multi-agent protocol around a reactive operational layer.

Decision intelligence

AHP-based supplier ranking makes weighting logic visible rather than hiding it behind a single score.

Operational immediacy

Three.js telemetry plus SSE exposes distributed system state as it changes.

Evidence integrity

Append-only SHA-256 linked blocks provide a verifiable audit trail and tamper sandbox.

3. Problem Discovery

Problem	What breaks	Product opportunity
Coordination friction	Supplier evaluation is subjective; multi-attribute trade-offs are difficult; negotiation is slow and siloed.	Unify selection, negotiation and operational state in one workflow.
Trust deficit	Delivery milestones, condition telemetry and SLA variance lack immutable record-keeping; audits are costly.	Create cryptographically verifiable evidence that can be checked on demand.
Dynamic performance	Static supplier reputation can conceal recent delivery defaults or changing performance.	Update reputation from settlement-linked evidence instead of stale scores.

4. Research Background & Objectives

Research methodology: technical secondary research, enterprise supply-chain pattern reviews, systems benchmarking, and architectural prototyping. The work intentionally avoids fabricated customer quotes, synthetic pilot revenues, or invented commercial metrics.

- Understand procurement trade-offs across price, lead time, reliability and ESG.
- Map RFQ and counter-offer friction, including deadlocks and runaway bidding loops.
- Determine the cryptographic proof artifacts auditors and compliance teams would need.
- Define telemetry requirements for real-time operational awareness.
- Identify safety guardrails for autonomous negotiation and failure handling.

HYPOTHESES — NOT STATISTICAL FINDINGS

AHP pairwise matrices may improve decision transparency; real-time telemetry may improve operator response; visible cryptographic invalidation may increase trust; telemetry-linked SLA logic may reduce dispute-resolution cycles.

5. Research Findings & Insight Framing

Observed / inferred pattern	Product insight	Opportunity
Procurement decisions involve non-price criteria.	Lowest price is not always the best business outcome.	Expose AHP trade-offs and decision rationale.
Negotiations can loop or deadlock.	Autonomy needs explicit boundaries and fallbacks.	Use round limits, reserve ceilings and timeouts.
Operational data becomes stale between polling cycles.	Operators need state changes, not just snapshots.	Push live SSE events into an observable console.
Historical records are difficult to trust after disputes.	Integrity should be demonstrable, not merely claimed.	Link blocks cryptographically and expose tamper verification.

6. Target Personas

Persona	Primary goals	Pain points	Core needs
Enterprise Procurement Lead	Optimize multi-vendor selection across cost, speed and ESG.	Manual RFQ cycles; opaque trade-offs.	Automated AHP scoring; transparent negotiation logs.
Supply Chain Operations Manager	Monitor global shipments and detect deviations in real time.	Fragmented tracking; delayed alerts.	3D telemetry; instant SSE alert feed.
Internal Compliance & Audit Officer	Verify validity, terms and payment releases.	Record alteration risk; manual checks.	SHA-256 chain verification; tamper auditing.

7. Jobs To Be Done

- When an unexpected component shortage occurs, autonomous agents should execute multi-round negotiations with pre-qualified vendors to balance delivery speed, price and quality.
- When proposals compete, strategic weights should be adjustable so the model reflects current business priorities.
- When global transit is active, real-time 3D spatial updates and telemetry alerts should make bottlenecks visible immediately.
- When contract milestones are audited, cryptographic proof should show that neither party modified transaction history.

8. End-to-End User Journey

1. Ingest	2. Formulate RFQ	3. Negotiate	4. Award	5. Commit	6. Settle
Live telemetry	Criteria + lot	Bounded agent rounds	AHP ranking	Hash-linked vault	SLA variance

Key risk progression: stale data → simplistic filtering → negotiation deadlock → opaque ranking → database tampering → manual SLA reconciliation.

PRODUCT STRATEGY · FOUR DESIGN PILLARS

01 CRYPTOGRAPHIC INTEGRITY

Evidence must be independently checkable.

02 EXPLAINABLE INTELLIGENCE

Show criteria, weights and ranking logic.

03 OPERATIONAL IMMEDIACY

Push system state instead of stale snapshots.

04 FRICTIONLESS EVALUATION

Let reviewers inspect the mechanics quickly.

MVP PRIORITIZATION

P0

3D telemetry ? SSE ? M2M negotiation ? AHP ? SHA-256 vault

P1

Tamper tool ? SLA / escrow settlement oracle

P2

Multi-tenant ERP connectors and production integrations

9. Competitive & Alternative Solutions

Alternative	Core strength	Critical limitation / AERIS advantage
Legacy ERP (SAP/Oracle)	Deep finance and vendor data.	Slow/monolithic; weak autonomous M2M negotiation and tamper demonstration.
Enterprise blockchain (Hyperledger)	Immutability and shared ledger patterns.	Heavy infrastructure for a portfolio prototype; no built-in operational 3D telemetry.
B2B procurement portals (Coupa/Ariba)	Structured purchasing and approvals.	RFQ remains relatively manual; limited dynamic trade-off modeling.
AERIS	Unified 3D visualization, AHP, bounded M2M negotiation and client-verifiable ledger.	Prototype stage; production ERP connectors and enterprise governance remain future work.

10. Product Goals & Non-Goals

GOALS

Working full-stack reactive prototype; transparent AHP across Price / Lead Time / Reliability / ESG; 3D nodes with SSE; interactive cryptographic ledger tamper sandbox.

NON-GOALS

Legally binding or high-throughput commercial trading; real fiat/gas; live proprietary ERP without adapters; fully autonomous financial disbursement without escrow boundaries.

11. System Architecture

USER / OPERATOR SURFACE	Three.js 3D Globe · Reactive DOM Console · Decision / Audit Views
REAL-TIME TRANSPORT	Server-Sent Events (SSE) → /events
RUNTIME	Node.js / Express / TypeScript state engine
PRODUCT CORES	Agent Negotiation Engine AHP Decision Analytics SLA & Escrow Oracle SHA-256 Audit Vault
BOUNDARIES	Human approval • reservation ceilings • timeouts / fallbacks • no real financial disbursement

12. Key Functional & Non-Functional Requirements

ID	Requirement	Priority
FR-01	Broadcast real-time system events over SSE /events.	P0
FR-02	Render interactive 3D globe with supplier nodes, latency and status.	P0
FR-03	Execute multi-round negotiations with dynamic price / lead-time concessions.	P0
FR-04	Rank suppliers using an Analytic Hierarchy Process matrix.	P0
FR-05	Commit completed negotiations to immutable blocks with valid parent hashes.	P0
FR-06	Corrupt block payloads interactively and test tamper detection.	P1
FR-07	Calculate SLA delivery variance and update supplier reputation.	P1
NFR	Deterministic crypto recalculation; <100 ms telemetry reflection target; safe agent timeout; portable cloud/container deployment.	NFR

13. Metrics & Measurement Framework

MetricKey indicatorENGINEERING / PRODUCT TRADE-OFFS

SSE OVER WEBSOCKETS

Simpler unidirectional telemetry and container deployment.

HYBRID CRYPTO SANDBOX

Fast portfolio proof without wallet / gas / testnet friction.

BOUNDED AUTONOMY

Reservation ceilings + round limits + safe fallbacks.

System integrity

Tamper Detection Latency < 50 ms

Tamper detection speed when ledger data is altered.

NORTH-STAR LOGIC

RESOLUTION

Eligible RFQ completes without manual override.

QUALITY

Decision remains mathematically consistent.

INTEGRITY

Altered history can be detected immediately.

14. What Was Validated vs. What Requires Enterprise Validation

Validated in prototype artifact	Requires real-world enterprise validation
Real-time 3D telemetry synchronized over SSE.	Scalability with thousands of concurrent supplier streams.
Multi-round negotiation convergence using bounded concession curves.	Supplier willingness to participate in autonomous pricing / negotiation.
AHP weighting, ranking and human approval state.	Decision usefulness, procurement policy fit and segregation of duties.
SHA-256 linked-block integrity and tamper detection.	Legal admissibility, key management, persistence and audit policy.
Prototype SLA variance / reputation logic.	Commercial ROI, dispute reduction and adoption at enterprise scale.

15. PM Retrospective & Key Takeaways

01	Algorithmic transparency builds trust	Autonomous agents should not operate as opaque heuristics; intermediate AHP weights make decisions inspectable.
02	Visualizing system state accelerates comprehension	SSE and cryptographic concepts become easier to understand when rendered as reactive system state.
03	Integrity must be demonstrable	The tampering sandbox is stronger evidence than simply asserting that the ledger is secure.
04	Prototyping validates product architecture	A focused full-stack build surfaces edge cases such as connection leaks and infinite negotiation loops early.

16. PM Decision Narrative · From Problem to Product

Decision	Why it mattered
1 — Make trade-offs explicit	AHP exposes weighting logic instead of reducing procurement to lowest price.
2 — Make autonomy bounded	Reservation ceilings, concession curves, round limits and fallbacks turn autonomy into a governed capability.
3 — Make system state observable	SSE plus 3D telemetry creates an operational surface for distributed behavior.
4 — Make integrity demonstrable	Hash-linked blocks and tamper testing let reviewers observe the security property.
5 — Protect scope	Production blockchain, live ERP integration and unrestricted autonomous settlement stay outside the prototype boundary.

17. Product Risk & Governance Matrix

Risk	Potential failure	Product control
Runaway negotiation	Agent keeps bidding without convergence.	Round limits, reservation boundaries, timeout/fallback.
Poor explainability	User cannot understand supplier ranking.	Visible AHP criteria, weights and scoring.
Stale operational state	Operator acts on old telemetry.	SSE stream and live state indicators.
Record tampering	Historical evidence is altered.	Linked SHA-256 hashes and tamper verification.
Unauthorized settlement	Autonomous system releases value incorrectly.	Human approval and bounded escrow logic.
Supplier performance drift	Static reputation becomes misleading.	Settlement-linked reputation updates.
Prototype overclaim	Portfolio artifact is mistaken for production.	Explicit validation and disclosure boundaries.

GOVERNANCE OPERATING MODEL

AUTONOMY

Allowed only within explicit commercial and policy bounds.

OBSERVABILITY

Material state changes remain visible to the operator.

AUTHORIZATION

Human approval stays the boundary for consequential value release.

VALIDATION BOUNDARY

PROTOTYPE PROOF

Mechanics, logic, observability and integrity are demonstrable.

ENTERPRISE PROOF

Scale, adoption, legal fit, ROI and production security need real validation.

18. Interview-Ready Case Study Summary

SITUATION Enterprise procurement workflows can be fragmented, manual and difficult to audit when multiple vendors, constraints and operational events interact.	TASK Design a product concept showing how autonomous agents could coordinate procurement while remaining explainable, observable and governed.	ACTION Defined users/JTBD, mapped the workflow, established product principles, prioritized MVP, selected TypeScript/SSE, introduced AHP, bounded autonomy and built an audit sandbox.
RESULT The prototype demonstrates the intended product mechanics and provides a concrete basis for usability, technical and enterprise validation. It does not claim commercial adoption, revenue or enterprise-scale performance.		

19. What Would Make the Case Study a Genuine 10/10

The next product-management iteration should add evidence rather than additional technology: 5–8 real target-user interviews; a research finding-to-decision traceability table; representative usability tests; measured task completion / time / error data; negotiation convergence measurements; cryptographic verification performance; and documented iterations based on observed user behavior.

RESEARCH	INSIGHT	DECISION	PRIORITIZATION	BUILD	TEST	MEASURE	ITERATE
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The strongest version of AERIS therefore tells a complete PM loop: **Research** → **Insight** → **Decision** → **Prioritization** → **Build** → **Test** → **Measure** → **Iterate**. Customer, revenue, ROI or enterprise-adoption claims should only be added when genuinely observed and documented.

20. Final Portfolio Positioning

AERIS — AI-Agent Enterprise Procurement Platform

Designed and prototyped an enterprise coordination layer enabling AI agents to discover suppliers, negotiate bounded procurement terms, evaluate multi-attribute trade-offs, obtain human authorization, and preserve cryptographically verifiable evidence.

BEST-FIT ROLES	Technical Product Manager • AI Product Manager • Platform Product Manager • Autonomous-Systems Product Roles
PM SIGNAL	Shows product thinking at the intersection of AI agents, distributed systems, decision science and enterprise governance.
LIVE PROTOTYPE	aeris-platform.onrender.com

Scope discipline: AERIS is a portfolio prototype. Its strongest claim is that the product mechanics, decision logic, observability and governance boundaries are demonstrable—not that enterprise-scale outcomes have already been proven.

PORTFOLIO EVIDENCE MAP			
DISCOVERY Problem framing ? personas ? JTBD ? alternatives	PRODUCT Goals ? non-goals ? MVP ? requirements ? roadmap	TECHNICAL Architecture ? SSE ? AHP ? cryptographic auditability	GOVERNANCE Risks ? bounded autonomy ? validation boundaries
ROLE FIT			
AI PRODUCT Agent behavior + explainability + guardrails	PLATFORM PRODUCT Coordination layer + integrations + observability	TECHNICAL PM Architecture trade-offs + requirements + delivery	